CM3604 – Deep Learning

Key information

Academic Year	**2026/2027**
Semester	**Semester 1**
Module Code	**CM3604**
Module Title	**Deep Learning**
Assessment Description	**Coursework – Group**
Deadline (date & time)	**11th November 2026 23:59**
48-Hour Automatic Extension Available	**Yes** *Exceptionally, indicate why the 48-hour automatic extension cannot apply, e.g. The 48-hour automatic extension is not available for practical exams.*
Submission	**Assessment Dropbox in the Module Study Area in CampusMoodle.**
Word Limit	**N/A**
Use of Generative Artificial Intelligence (GenAI) **See** Using Gen AI at RGU	**3 Editing**
Module Coordinator	**Prasan Yapa**

What knowledge and/or skills will I develop by undertaking the assessment?

The primary goal with this task is to give you hands-on experience analyzing outputs from large language models (LLMs). You will understand what sorts of non-factual outputs LLMs can generate, and what is involved in the process of verifying those outputs against Wikipedia. In addition, you will conduct some error analysis of the mistakes your fact-checking model makes.
On successful completion of the assessment the student will be able to: 1. Design and build a neural model by selecting relevant functions and parameters using a state-of-the-art python framework. 2. Implement, test, and transfer deep architectures for a given data science application.
Please also refer to the Module Descriptor, available from the module Moodle study area.

What is expected of me in this assessment?

Task(s) – content

Please use Python 3.10 and the specified version of packages listed in `requirements.txt`. The dataset and experimental setup used in this project are based on FActScore. FActScore focuses on identifying factual errors in biographies generated by ChatGPT. Its approach involves breaking down generated claims into individual facts and searching Wikipedia for evidence to verify whether those facts are supported. They conduct human annotation of these steps to understand the performance of systems in different parts of this pipeline. The `facts` you are given are propositions extracted from ChatGPT outputs. These are simple natural language sentences. Although FActScore explores different ways of producing these facts from biographies, we use their *human-annotated* facts for our analysis, to remove any potential errors in this step. These facts are then validated against `passages` that are retrieved by Wikipedia. Each fact is hand-labeled by humans with either "S" for supported, "NS" for not supported, or "IR" for irrelevant.

The framework code you are given consists of two files. `factchecking_main.py` loads the data, instantiates the `FactChecker` model to use, executes it on the data, and reports accuracy, like the driver classes in similar projects. **You will not be modifying this file**. `factcheck.py` contains `FactExample`, `EntailmentModel`, and the `FactChecker` classes you will implement. The raw fact data is located at `data/dev_labeled_ChatGPT.jsonl` and the retrieved passages are in `passages_bm25_ChatGPT_humfacts.jsonl`.

In the provided Python code, `FactExample` stores the examples themselves. These consist of a fact (a string), passages (a list of dictionaries consisting of a title and text), and the label. Note that the passages themselves are not tokenized or preprocessed in any way.

Task(s) – format

Part 1: Word Overlap (40 Marks)

In this first part, you will explore for fact-checking in bag-of-words overlap. A supported fact might be expected to have high unigram overlap with a passage that supports it, and low overlap with passages that don't support it. You should implement a method that predicts supported vs. not supported based on this criterion. You will have to decide on several details of your implementation. *How do you want to handle tokenization*? *Do you want to stem*? *Do you want to remove stop words*?

Your method should eventually compute a real-valued score that you use to make your classification decision. This could be a cosine similarity of TF-IDF vectors, Jaccard similarity of word sets, metrics based on precision/recall, existing lexical metrics like ROUGE or BLEU, or anything similar that computes a discrete word overlap score.

Your method should get at least 75% accuracy on the dataset to receive full credit. Try tuning the preprocessing method as well as changing the classification threshold to achieve the best performance. You can tune this threshold on the development set to optimize for accuracy. It

would be best practice to tune this on a separate set, but we won't have separate dev and test sets here.

Part 2: Textual Entailment (40 Marks)

In this section, you will explore a method that goes beyond the surface form of the words. The most appropriate type of system explored in prior work is textual entailment. These systems should theoretically be able to decide whether a fact (the "hypothesis") is logically entailed by a source document (the "premise").

For this part, you will use a pre-trained model, specifically a DeBERTa-v3 base model. This is a pre-trained instance of DeBERTa fine-tuned on the MNLI, FEVER, and ANLI. This model can take a premise-hypothesis pair and return a decision: entailment, neutral, or contradiction. Your task is to use this information to determine supported vs. not supported for the fact.

Getting started Run: `python factchecking_main.py --mode entailment --cuda`

The optional `--cuda` flag to do the entailment model queries on a GPU. **You should do this only if you have Pytorch with CUDA installed and a GPU enabled**.

The model itself is loaded in `factchecking_main.py`. In `factcheck.py`, the `EntailmentModel` class includes most of the boilerplate code needed to do inference with this model. However, we stop short of mapping logits to an actual prediction. You will have to decide what strategy makes the most sense for extracting a binary S/NS decision from a three-class entailment/neutral/contradiction decision. You can either use the discrete entailment decision, or you can derive a decision based on a threshold of one of the entailment/neutral/contradiction probabilities.

If your implementation uses the entailment classifier discretely, then the sentence can be considered entailed if it is entailed by any part of any of the passages. If your implementation uses entailment scores and sets a threshold, then the score for that fact should be the max over the entailment scores returned by running on any example in the passage. **Your method should get at least 83% accuracy on the dataset to receive full credit**.

Running the entailment model on all the data may be slow. To speed things up, you can implement pruning based on the word overlap method. If there is very low word overlap, then entailment is extremely unlikely. You can discard examples that fail to meet a low word overlap threshold, then run entailment on the remaining instances. Your code must run without causing out-of-memory issues.

Part 3: Error Analysis (20 Marks)

Part 3 involves a written submission for an error analysis on the results of the entailment model. There are two broad types of errors you should investigate. False positives are cases where the model predicted "supported" but the ground truth label is "not supported". False negatives are the opposite, where the model predicted "not supported" but the ground truth label is "supported".

You should manually examine 10 errors of each type (10 false positives and 10 false negatives), or as many errors as your system makes if it does not make 10. You should categorize them into a set of a few fine-grained categories that you see as suitable. You should aim for 2-4 fine-grained error categories collectively between false positives and false negatives; you do not need to come up with many categories, but it's important to analyze the data accurately. **These categories should not just be "false positive" or "false negative"** but should identify common language or other patterns in the examples where the model was wrong. To receive full credit for this part, your fine-grained categories should be non-trivial and the statistics and examples in your writeup should support your categorizations. In your writeup, include:

1. Brief definitions of the fine-grained error categories used in your analysis (1-2 sentences per category).
2. Aggregate statistics about the errors you examined: how many errors were of each fine-grained type?
3. Three examples discussed in more detail. For each of these, include: (a) the text of the example itself; (b) the ground truth label; (c) your model's predicted label; (d) the fine-grained error type you gave it; (e) 1-3 sentences describing why you think the fine-grained error type label applies.

Submission: You should submit the following files separately.

1. A PDF of your written submission.
2. `factcheck.py` for the first two parts.

Make sure that the following commands work before you submit:

```
python factchecking_main.py --mode word_overlap
```

```
python factchecking_main.py --mode entailment
```

These commands should run without errors.

How will I be graded?

A grade will be provided for each criterion on the feedback grid which is specific to the assessment. The overall grade for the assessment will be calculated using the algorithm below.	
A	At least 50% of the feedback grid to be at Grade A, at least 75% of the feedback grid to be at Grade B or better, and normally 100% of the feedback grid to be at Grade C or better.
B	At least 50% of the feedback grid to be at Grade B or better, at least 75% of the feedback grid to be at Grade C or better, and normally 100% of the feedback grid to be at Grade D or better.
C	At least 50% of the feedback grid to be at Grade C or better, and at least 75% of the feedback grid to be at Grade D or better.
D	At least 50% of the feedback grid to be at Grade D or better, and at least 75% of the feedback grid to be at Grade E or better.
E	At least 50% of the feedback grid to be at Grade E or better.
F	Failing to achieve at least 50% of the feedback grid to be at Grade E or better.
NS	Non-submission.

Feedback Grid

GRADE	A	B	C	D	E	F
DEFINITION/ CRITERIA (WEIGHTING)	**EXCELLENT** Outstanding Performance	**COMMENDABLE/VERY GOOD** Meritorious Performance	**GOOD** Highly Competent Performance	**SATISFACTORY** Competent Performance	**BORDERLINE FAIL**	**UNSATISFACTORY** Fail
Word Overlap (40%) **Grade:** ☐	Implements a complete, robust, and technically sound bag-of-words overlap approach for support prediction. Makes well-justified preprocessing decisions (tokenization, stemming/lemmatization, stop-word handling) supported by experiments or literature. Correctly implements one or more appropriate similarity/scoring methods (e.g., TF-IDF cosine similarity, Jaccard similarity, precision/recall overlap, ROUGE, BLEU) with clear mathematical understanding. Systematically tunes preprocessing choices and optimizes classification thresholds using appropriate validation procedures. Demonstrates critical evaluation of alternative methods, provides comprehensive experimental analysis, and justifies all implementation decisions. Implementation is accurate, efficient, and reproducible.	Implements a correct and effective bag-of-words overlap classifier with appropriate preprocessing. Uses suitable similarity or overlap metrics correctly and demonstrates good understanding of their application. Performs threshold optimization and some preprocessing tuning with reasonable justification. Experimental evaluation is thorough, although analysis of alternatives or parameter sensitivity may lack depth. Implementation is correct with only minor limitations.	Implements a working bag-of-words overlap classifier that correctly predicts supported vs. not supported in most cases. Applies basic preprocessing techniques appropriately but with limited justification or experimentation. Uses an appropriate scoring metric (e.g., TF-IDF similarity or word overlap) correctly. Performs limited threshold tuning or preprocessing optimization. Evaluation demonstrates functionality but lacks detailed analysis or comparison between alternative approaches. Minor implementation issues may exist but do not significantly affect performance.	Implements a basic classifier that functions but demonstrates only a limited understanding of preprocessing and similarity measures. Uses simple overlap or similarity metrics with minimal justification. Threshold selection is simplistic or manually chosen without systematic optimization. Little or no tuning of preprocessing methods is performed. Evaluation is basic and identifies only obvious strengths or weaknesses. Implementation may contain inefficiencies or minor methodological errors.	Produces a partially functioning implementation with significant methodological weaknesses. Preprocessing decisions are incomplete, inconsistent, or inappropriate. Similarity or scoring methods are incorrectly implemented or poorly justified. Threshold selection is arbitrary or absent. Minimal evaluation is provided, with little evidence of testing or optimization. Important components are missing or contain substantial errors that limit predictive performance.	Fails to implement a working bag-of-words overlap classifier or produces an implementation with fundamental technical errors. Preprocessing is absent or inappropriate. Similarity metrics are missing or incorrectly implemented. No meaningful threshold selection, optimization, or evaluation is performed. Demonstrates little understanding of the required methods, with major omissions or incorrect application of core concepts.

GRADE	A	B	C	D	E	F
DEFINITION/ CRITERIA (WEIGHTING)	**EXCELLENT** Outstanding Performance	**COMMENDABLE/VERY GOOD** Meritorious Performance	**GOOD** Highly Competent Performance	**SATISFACTORY** Competent Performance	**BORDERLINE FAIL**	**UNSATISFACTORY** Fail
Textual Entailment (40%) **Grade:** ☐	Implements a robust and well-justified strategy for converting three-class entailment/neutral/contradiction predictions into binary Supported (S) / Not Supported (NS) decisions. Critically evaluates alternative strategies and provides clear justification supported by experiments. Correctly preprocesses passages through effective sentence splitting and noise removal. Appropriately incorporates pruning based on low word-overlap thresholds, demonstrating measurable improvements in efficiency and/or predictive performance. The implementation is computationally efficient, scalable, and executes reliably without out-of-memory issues. Experimental evaluation thoroughly analyses the impact of each design choice, including threshold selection, pruning, and preprocessing.	Implements an effective binary decision strategy from three-class entailment outputs with appropriate justification. Uses either discrete entailment decisions or threshold-based classification correctly. Preprocesses passages into sentences with suitable cleaning techniques. Incorporates pruning effectively with reasonable threshold selection and demonstrates its benefit through evaluation. The implementation runs efficiently without memory issues, although analysis of alternative strategies or optimization could be more comprehensive.	Implements a correct binary classification strategy that functions reliably, although justification for the chosen approach is limited. Sentence splitting and basic cleaning are correctly applied. Pruning is implemented but may be simplistic, minimally tuned, or only partially evaluated. The system generally executes without memory problems. Experimental evaluation demonstrates functionality but provides limited discussion of the effects of preprocessing, pruning, or threshold choices.	Implements a basic approach for deriving binary decisions from three-class predictions but with limited methodological understanding or justification. Sentence preprocessing is present but may be incomplete or inconsistent. Pruning is absent, minimally implemented, or applied without appropriate rationale. Threshold selection is simplistic or arbitrary. The implementation generally runs but may exhibit inefficiencies or occasional memory-related limitations. Evaluation is basic and lacks detailed analysis of design decisions.	Produces a partially functioning implementation with significant weaknesses. The mapping from three-class predictions to binary decisions is poorly justified or incorrectly implemented. Sentence splitting or text cleaning is incomplete or ineffective. Pruning is missing or incorrectly applied. Memory management is poor, with inefficient processing or execution issues on larger datasets. Evaluation is minimal and provides little evidence that implementation choices were considered or validated.	Fails to implement a functioning strategy for deriving binary Supported/Not Supported decisions from three-class entailment outputs. Preprocessing is absent or fundamentally incorrect. Sentence splitting, cleaning, or pruning are missing or non-functional. The implementation suffers from serious computational or memory issues that prevent successful execution. Little or no evaluation is provided, demonstrating insufficient understanding of the required methods and their implementation.
Error Analysis						

GRADE	A	B	C	D	E	F
DEFINITION/ CRITERIA (WEIGHTING)	**EXCELLENT** Outstanding Performance	**COMMENDABLE/VERY GOOD** Meritorious Performance	**GOOD** Highly Competent Performance	**SATISFACTORY** Competent Performance	**BORDERLINE FAIL**	**UNSATISFACTORY** Fail
(20%) **Grade:** ☐	Conducts a thorough and systematic error analysis by examining 10 false positives and 10 false negatives, or all available errors where fewer exist. Develops meaningful, fine-grained error categories that identify recurring linguistic, semantic, or structural patterns rather than simply listing mistakes. Provides clear aggregate statistics summarizing the distribution of error types and identifies key trends. Each error category is supported by concise, insightful explanations (1–3 sentences) that accurately justify why the examples belong to that category. Demonstrates critical reflection on the limitations of the model and discusses implications for future improvements.	Examines the required false positives and false negatives (or all available errors) and identifies several meaningful recurring error categories. Categories go beyond simple descriptions of incorrect predictions and capture common patterns in the data. Provides appropriate aggregate statistics and clear justifications for each category. Analysis demonstrates good understanding of model behavior, although discussion of broader implications or category definitions may lack depth.	Examines the required errors and groups most into reasonable categories, although some categories may be broad, overlapping, or only partially capture recurring patterns. Aggregate statistics are provided but may be limited in detail. Justifications generally explain why examples belong to each category but may be brief or lack insight. The analysis demonstrates a sound understanding of common model errors but offers limited critical evaluation.	Examines a reasonable number of errors but with limited depth or consistency. Categories tend to describe isolated mistakes rather than identifying meaningful recurring language or structural patterns. Aggregate statistics are basic or incomplete. Explanations for category assignments are present but superficial or only partially justified. Limited reflection is provided on what the errors reveal about model performance.	Examines only a subset of the required errors or provides a largely descriptive list of mistakes without meaningful categorization. Error categories are poorly defined, inconsistent, or fail to identify common patterns. Aggregate statistics are missing or incomplete. Explanations are minimal, unclear, or unsupported. Demonstrates limited understanding of systematic error analysis.	Fails to perform a meaningful error analysis. Required false positives and false negatives are largely absent or not examined. Errors are not categorized, or categories are incorrect or trivial. Aggregate statistics are missing. Explanations are absent or demonstrate little understanding of why the model failed. The submission provides no useful insight into model behavior or avenues for improvement.

Coursework received late, without valid reason, shall not be accepted and shall receive no grade, but shall count as one of the assessment attempts.

What else is important to my assessment?

What is the Assessment Word Limit Statement?

It is important that you adhere to the Word Limit specified above. The Assessment Word Limit Statement can be found in Appendix 3 of the RGU Assessment Policy. It provides detail on the purpose, setting and implementation of wordage limits; lists what is included and excluded from the word count; and the penalty for exceeding the word count.

What's included in the word count?

The table below lists the constituent parts which are included and excluded from the word limit of a Coursework; more detail can be found in the full Assessment Word Limit Statement. Images will not be allowed as a mechanism to circumvent the word count.

Excluded	Included
• Cover or Title Page • Executive Summary (Reports) or Abstract • Contents Page • List of Abbreviations / Acronyms • List of Tables / Figures • Tables – mainly numeric content • Figures • Reference List and/or Bibliography • Appendices • Glossary	• Main Text e.g. Introduction, Literature Review, Methodology, Results, Discussion, Analysis, Conclusions, and Recommendations • Headings and subheadings • In-text citations • Footnotes (relating to in-text footnote numbers) • Quotes and quotations written within "…" • Tables – mainly text content

What are the penalties?

The grade for the submission will be reduced to the next lowest grade if:

- The word count of the submission is above the specified word limit by more than 10%.
- The submission contains an excessive use of text within Tables or Footnotes.

What are the categories of GenAI use?

The table below describes the different categories of GenAI use in assessment.

Category	Description *
1 **No AI**	GenAI use is not authorised. • You must not use GenAI at any point to complete the assessment.
2 **Ideas**	You may use GenAI to plan, research and generate ideas for the content and structure, or on how to improve the work. • This may be optional or a requirement of the assessment. • No GenAI content is allowed in the final submission.
3 **Editing**	You may use GenAI to improve the communication and appearance of your work, i.e. to refine, edit and enhance the language or presentation of the original work (e.g. proofreading; image editing). • This may be optional or a requirement of the assessment. • No new ideas can be created. • English is the language of assessment at RGU, therefore submitted work must not include text produced using translation tools (false authorship).
4 **Tasks**	You can use GenAI to complete certain elements of the assessment task. • This may be optional or a requirement of the assessment.
5 **Full AI**	You can use GenAI throughout the assessment. • This may be optional or a requirement of the assessment. • English is the language of assessment at RGU, therefore submitted work must not include text produced using translation tools (false authorship).

* For all categories, please see the Academic Integrity in Practice resource which contains guidance on Document Control for maintaining academic integrity. Keep a copy of all coursework and any material demonstrating how the coursework was produced until after receipt of a confirmed result from the Assessment Board and the period for submission of academic appeals has elapsed.

Citing and referencing use of GenAI

RGU Library has guidance for citing and referencing use of GenAI in the style of Harvard, Vancouver, APA and OSCOLA.

Acknowledging use of GenAI

If you have to acknowledge GenAI use, you can use this template unless otherwise specified:

Delete as appropriate:

- I have not used GenAI to help me complete my assessment.
- I acknowledge use of [insert name of GenAI tool(s)] from [url of GenAI tool(s)] to help me complete my assessment in the following ways:

  Delete as appropriate:

  - To help me search for literature related to my assessment.
  - To help me generate ideas and/or potential topics for my assessment.
  - To help me create a structured outline for my assessment.
  - To help me proofread my assessment.
  - To help me improve the wording and/or phrasing in my assessment.
  - To help me edit the length of my assessment.
  - To help in the following ways not covered by the above [student should enter free text].
- I entered the following prompts on [insert date]: [insert prompts] and [insert description of how the generated content was used].

Academic Integrity

For information on academic integrity, please see the Moodle page Academic Integrity and the resource Academic Integrity in Practice which contains guidance on Document Control for maintaining academic integrity. Categories of academic misconduct are defined in RGU Academic Regulation A3-2 Student Conduct Procedure and include the definitions below.

What is false authorship?

"The University defines this as the practice of submitting work where the student is not the author of that work. The 'false authorship' may relate to the student engaging with a third party and/or software tool to complete an assessment, either in part or whole. This may include work produced by, but not attributed to: another student, an essay mill, a family member or friend, a tutoring service or the unauthorised use of Artificial Intelligence (AI) software. It may also include payment, or other favours, though this will not always be the case. It may relate to any form of assessment or conducting research".

What is plagiarism?

Plagiarism is “the practice of presenting the thoughts, writings or other output of another or others as original, without acknowledgement of their source(s) at the point of their use in the student’s work. All materials including text, data, diagrams or other illustrations used to support a piece of work, whether from a printed publication or from electronic media, should be appropriately identified and referenced and should not normally be copied directly unless as an acknowledged quotation. Text, opinions or ideas translated into the words of the individual student should in all cases acknowledge the original source”.

What is collusion?

Collusion is defined as “two or more students working together, without the prior authorisation of the Course Leader, tutor or supervisor, to produce the same piece of work, and then attempting to present this work as their own”.

What if I’m unable to submit?

- The University operates a Fit to Sit Policy which means that if you undertake an assessment then you are declaring yourself well enough to do so.
- If you require an extension due to extenuating circumstances, you should complete and submit a Coursework Extension Form. This form is available on the RGU Student and Applicant Forms page.
- All extension requests must be submitted a minimum of 24 hours in advance of the specified deadline and extensions will be provided for a maximum of 5 working days from the specified deadline.
- You are expected to submit your work by the specified deadline. The University does operate a 48-Hour Automatic Extension which is available to you in the event you need it. The purpose of the automatic extension is to support you with minor illness, unexpected or adverse events that might be impacting your ability to submit your coursework by the submission date. These do not constitute extenuating circumstances.
- There may be limited instances where the 48-Hour automatic extension cannot apply and the Coursework Assessment Brief will state if this is the case in the ‘48-Hour Automatic Extension Available’ box, therefore please check this Assessment Brief carefully. Please also refer to Regulation A4: Assessment and Recommendations of Assessment Boards and the Fit to Sit Policy: Extenuating Circumstances.
- Further support is available from your Course Leader.

What additional support is available?

- RGU Study Skills provide advice and guidance on academic writing, study skills, maths and statistics and basic IT.
- RGU Library guidance on referencing and citing.
- The Inclusion Centre: Disability & Dyslexia.
- Your Module Coordinator and Course Leader can also provide support.

What are the University rules on assessment?

The University Regulation 'A4: Assessment and Recommendations of Assessment Boards' sets out important information about assessment and how it is conducted across the University.